13F Intelligence Platform — Developer Spec
Fund Ranking + Stock Ranking Pipeline

Overview

Transform ingested 13F SEC filing data into two ranked outputs:
1. **Fund Rankings** — which small, concentrated funds have the best long-term track records
2. **Stock Rankings** — which stocks those top funds are most convicted on right now
Both outputs are displayed on the website: a raw version and a filtered/curated version.

Scoring methodology is framed publicly as "Holding Period Return Simulation" — it measures whether a fund's stock picks at each filing date appreciated over 3 years on a static hold basis. This isolates selection skill from trading skill and is not a measure of actual fund performance.

Data Assumptions (What We Have)

The following data is already ingested and available:

Table	Fields
`holdings`	fund_id, quarter_date, ticker, cusip, position_value_usd, share_count
`funds`	fund_id, fund_name, first_filing_date
`portfolios`	fund_id, quarter_date, total_portfolio_value_usd
`prices`	ticker, date, close_price, adjusted_close_price

Notes for developer:
- Some identifiers in holdings are CUSIPs not tickers — need a CUSIP→ticker mapping table or resolution step before scoring
- Options/derivatives positions exist in the data — exclude these from all calculations (equity positions only)

- Quarter dates follow 13F filing cadence: March 31, June 30, September 30, December 31

PIPELINE 1: FUND RANKING

Stage 1 — Weeding

Define a single global constant at the top of the pipeline:

```
current_quarter_date = most recent 13F quarter end date (e.g. 2024-12-31)
```

This value is applied identically to every fund. Do not use each fund's own most recent quarter — all funds must be evaluated against the same period.

```
FILTER OUT if ANY of:
 MAX(position_value_usd) > 100,000,000     -- any single position over $100M
 COUNT(ticker) > 30                         -- more than 30 equity positions
 (TODAY - first_filing_date) < 5 years      -- less than 5 years of filing history
 last_filing_date < current_quarter_date    -- did not file in the most recent quarter
```

Store result as `fund_eligibility` table:

```
fund_id | eligible (bool) | fail_reason (string or null)
```

fail_reason values:

```
"position_too_large"    -- single position exceeded $100M
"too_many_positions"    -- more than 30 equity positions
"insufficient_history"  -- less than 5 years of filing history
"inactive"              -- did not file in most recent quarter
```

```
null                    -- fund passed all filters
```

All subsequent pipeline stages only process eligible funds.

Stage 2 — Forward Return Join

For each holding in `holdings` (eligible funds only):

```
3yr_return(ticker, quarter_date) =
 (adjusted_close_price(ticker, quarter_date + 3 years) - adjusted_close_price(ticker, quarter_date))
 / adjusted_close_price(ticker, quarter_date)
```

IMPORTANT — Look-ahead bias rule:
When scoring any historical quarter, only price data within that quarter's
exact 3-year window may be used. No prices from after the forward date
may be used in that quarter's calculation.

Example: Q3 2021 scoring uses prices between 2021-09-30 and 2024-09-30 only.

If the 3-year forward date price is unavailable, use the last available
adjusted price before that date instead. This handles delistings,
take-privates, and acquisitions in one rule — the market price at exit
naturally reflects the outcome (near zero for bankruptcy, at premium
for acquisitions).

Edge cases — handle explicitly:

Situation	Treatment
Ticker delisted before 3yr mark	Use last available adjusted price as forward price. Bankruptcy approaches zero naturally. Acquisition premium reflected naturally in final traded price.

Price data missing entirely	Flag as NULL, exclude from that quarter's QPS, renormalize remaining position weights to 100%
Corporate action (merger/acquisition)	Covered by last-price rule — acquisition price will be the final traded price before delisting
Spinoff	Record return on the original ticker up to the spinoff date. Then track both successor tickers weighted by their relative value at spinoff. Combined return = weighted average of both legs. If successor ticker data is unavailable, flag NULL and exclude.
CUSIP not resolved to ticker	Flag as unresolvable, exclude from scoring

Store as `holding_returns`:
```
fund_id | quarter_date | ticker | position_value_usd | 3yr_return | data_quality_flag
```

data_quality_flag values:
```
"clean"             -- full 3yr price data available
"last_price"        -- delisted/acquired, used last available price
"spinoff"           -- spinoff handled, both legs tracked
"spinoff_partial"   -- spinoff, one leg missing, partial return used
"null_excluded"     -- excluded from QPS, weights renormalized
"cusip_unresolved" -- excluded, CUSIP could not be mapped
```

Stage 3 — Quarterly Performance Score (QPS)

For each fund x quarter combination:

```
weight(i) = position_value(i) / SUM(position_value) for all positions in that quarter

raw_QPS(fund, quarter) = SUM [ weight(i) x 3yr_return(i) ]
```

benchmark_return(quarter) = S&P 500 total return from quarter_date to quarter_date + 3 years

excess_QPS(fund, quarter) = raw_QPS(fund, quarter) - benchmark_return(quarter)
```

Excess QPS is the primary scoring metric. A positive value means the fund's
portfolio beat the S&P 500 over that 3-year window. A negative value means
it underperformed.

Only include quarters where 3-year forward data exists (i.e. filed at least 3 years ago).

Store as `fund_quarterly_scores`:
```
fund_id | quarter_date | qps_raw | qps_excess | benchmark_return | positions_included | positions_excluded_null
```

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### Stage 4 — Time-Weighted Score (TWS)

For each fund, across all eligible quarters:

IMPORTANT: Use price at time of filing, NOT price at quarter end.

Minimum requirements before TWS is calculated:
```
 - Fund must have passed weeding with 5+ years of filing history (see Stage 1)
 - Fund must have at least 6 scoreable quarters (quarters where 3yr forward data exists)
 - If either condition is not met, fund is excluded from scoring entirely
   and marked ineligible in fund_eligibility table with
   fail_reason "insufficient_scoreable_quarters"
```

```

```
lambda = 0.85

w(t) = lambda ^ (quarters_from_most_recent)
 -- most recent quarter: w = 1.0
 -- 1 quarter back:      w = 0.85
 -- 2 quarters back:     w = 0.72
 -- etc.

TWS(fund) = SUM [ w(t) x excess_QPS(t) ] / SUM [ w(t) ]
```

One-hit wonder check:
```
best_quarter_contribution = MAX(w(t) x excess_QPS(t)) / SUM(w(t) x excess_QPS(t))

IF best_quarter_contribution > 0.50:
 flag fund as one_hit_wonder = true
 apply discount: TWS = TWS x 0.75
 record flag in fund_tws table
```

Store as `fund_tws`:
```
fund_id
tws
quarters_scored
oldest_quarter_included
one_hit_wonder_flag          -- bool
best_quarter_contribution    -- decimal, % of TWS from single best quarter
```

Stage 5 — Turnover Rate

For each fund, for each pair of consecutive quarters:

```
turnover(t) = COUNT(tickers in t-1 NOT in t) / COUNT(tickers in t-1)
```

```
avg_turnover(fund) = MEAN( turnover(t) for all consecutive quarter pairs )

turnover_multiplier = CLAMP( 1 - (avg_turnover x 0.5), min=0.5, max=1.0 )
```

Examples:
- Fund replacing 10% per quarter → multiplier = 0.95
- Fund replacing 40% per quarter → multiplier = 0.80
- Fund replacing 100% per quarter → multiplier capped at 0.50

Store as `fund_turnover`:
```
fund_id | avg_turnover_rate | turnover_multiplier | quarter_pairs_measured
```

Stage 6 — Consistency Score

```
raw_consistency(fund) = STDEV( excess_QPS(t) for all quarters )

-- Lower stdev = more consistent = better score
-- Rank all eligible funds by raw_consistency ascending
-- Convert to percentile (0 to 1), where 1 = most consistent

consistency_percentile(fund) = 1 - PERCENT_RANK(raw_consistency)
```

Must run across all funds simultaneously — percentile is relative, not absolute.

Known limitation: consistency score measures outcome consistency, not skill consistency.
Quarters with extreme market-wide events (e.g. COVID drawdown) may unfairly penalize
funds. Consider flagging macro shock quarters for optional exclusion in a future iteration.

Store as `fund_consistency`:
```
```

```
fund_id | qps_stdev | consistency_score (0-1)
```

Stage 7 — Composite Fund Score

```
raw_composite = (TWS x turnover_multiplier x 0.70) + (consistency_score x 0.30)

-- Normalize to 0-100 across all eligible funds
final_score = ( (raw_composite - MIN) / (MAX - MIN) ) x 100
```

Must run after all funds complete Stages 2-6.

Final output table `fund_rankings`:
```
fund_id
fund_name
rank                          -- 1 = best
final_score                   -- 0 to 100
tws_raw                       -- raw time-weighted excess return (decimal)
avg_turnover_rate             -- 0 to 1
turnover_multiplier           -- 0.5 to 1.0
consistency_score             -- 0 to 1
one_hit_wonder_flag           -- bool
best_quarter_contribution     -- decimal, % of TWS from single best quarter
quarters_of_data              -- how many quarters were scored
avg_position_count            -- average positions per quarter
avg_aum                       -- average total portfolio value
eligible                      -- true
fail_reason                   -- null for eligible funds
```

PIPELINE 2: STOCK RANKING

Stage 1 — Restrict Universe

Only include stocks held by funds in the top 50% of fund_rankings (by final_score).

```
qualifying_funds = fund_rankings WHERE rank <= (total_eligible_funds / 2)

stock_universe = DISTINCT tickers held by qualifying_funds in most recent quarter
```

Stage 2 — Per-Stock Signal Aggregation

For each stock in the universe, aggregate across all qualifying funds holding it:

Signal 1: Weighted Fund Conviction
```
-- How strong are the funds backing this stock, weighted by their rank score
fund_conviction(stock) = SUM [ fund_score(f) x weight(stock, f) ] / SUM [ fund_score(f) ]

where weight(stock, f) = position_value(stock in fund f) / total_portfolio_value(fund f)
```

Signal 2: Fund Count
```
holder_count(stock) = COUNT(qualifying funds holding this stock)
```

Signal 3: Net Positioning Change
```
-- Aggregate buying/selling pressure across qualifying funds QoQ
net_change(stock) = SUM across funds of:
  IF new position:      +position_value
  IF increased:         +increase_in_value
  IF decreased:         -decrease_in_value
  IF exited:            -prior_position_value
```

-- Normalize by total universe AUM to get a relative signal
net_change_pct = net_change / SUM(total_portfolio_value for qualifying funds)
```

**Signal 4: Average Relative Size**
```
-- How large is this position relative to each fund's total portfolio
avg_relative_size(stock) = MEAN [position_value(stock,f) / total_portfolio_value(f)]
 across all qualifying funds holding it
```

**Signal 5: Holding Tenure**
```
-- How many consecutive quarters has each qualifying fund held this stock
-- Funds that have held and not exited across multiple quarters signal stronger conviction

avg_tenure(stock) = MEAN [consecutive_quarters_held(stock, f)]
 across all qualifying funds holding it

-- consecutive_quarters_held resets to 1 if a fund exits and re-enters a position
```

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### Stage 3 — Fundamental Variables

Pull for each stock at the time of the most recent quarter filing:

| Variable | Source | Null Handling |
|---|---|---|
| Market cap | Price x shares outstanding | Required — exclude stock if missing |
| 52-week high/low | Price history | Calculate from prices table. 52wk_range_position = (price - 52wk_low) / (52wk_high - 52wk_low). If full 52 weeks unavailable: 4+ weeks of data: use available
```

history, set 52wk_partial = 1. Less than 4 weeks: set 52wk_range_position = NULL, set 52wk_partial = 1. Never exclude the stock entirely due to missing range data alone. Add 52wk_partial as a feature in the regression (0/1 dummy) so the model can discount partial range signals. |
P/E ratio	Earnings data	If negative or N/A → create dummy flag pe_available (0/1), use 0 for P/E in regression
Sector	Sector classification table	Required — use as dummy variable
Gross margin %	Financial statements	If missing → flag NULL, exclude from that variable only

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Stage 4 — Stock Scoring (Regression)

Run a weighted linear regression predicting 3yr_return using the signals above.

Training set: All historical stock x quarter observations where 3yr forward data exists,
limited to stocks held by top-50% funds at those historical quarters.

Features (X):
```
fund_conviction          -- weighted avg fund score of holders
holder_count             -- number of qualifying funds holding
net_change_pct           -- aggregate positioning change
avg_relative_size        -- average portfolio weight
avg_tenure               -- average consecutive quarters held across holding funds
log(market_cap)          -- log-transform to normalize
52wk_range_position      -- 0 to 1 (NULL if less than 4 weeks of data)
52wk_partial             -- 0/1 dummy, 1 = less than 52 weeks of price history
pe_ratio                 -- 0 if N/A
pe_available             -- 1/0 dummy
sector_dummies           -- one-hot encoded sectors
```

Target (Y):

```
3yr_return (from holding_returns table, same methodology as fund pipeline)
```

Sector adjustment:
After regression, compute sector-adjusted score:
```
sector_adjusted_score = raw_score - sector_mean_score
```
This prevents sector bias from dominating the rankings.

Stage 5 — Confidence Score

Confidence is a composite of five signals, not a simple fund count threshold.
It measures how strongly the data supports the stock's ranking, independent
of the rank itself.

Components (each normalized 0-1 before weighting):

1. Weighted Holder Score (30%)
```
-- Fund count weighted by fund rank, not raw count
-- A stock held by 3 top-10 funds scores higher than one held
   by 6 funds ranked 80-100

weighted_holder_score =
  SUM [ fund_score(f) for all qualifying funds holding stock ]
  / MAX possible score across all stocks in universe
  -- normalize to 0-1 across all stocks
```

2. Average Tenure Score (25%)
```
-- How many consecutive quarters has each holding fund held this stock
-- Funds had multiple opportunities to exit and didn't — stronger signal

```

avg_tenure_score =
 MEAN [consecutive_quarters_held(stock, f)]
 across all qualifying funds holding stock
 -- normalize to 0-1 across all stocks
```

**3. Average Relative Size (20%)**
```
-- How large is this position relative to each fund's total portfolio
-- A stock representing 15% of a fund is a different signal than 0.5%

avg_relative_size =
 MEAN [position_value(stock, f) / total_portfolio_value(f)]
 across all qualifying funds holding stock
 -- normalize to 0-1 across all stocks
```

**4. Direction Agreement (15%)**
```
-- Are qualifying funds agreeing on buy vs sell direction
-- Mixed conviction (half buying, half selling) is a weak signal

buyers = COUNT(funds where net_change > 0)
sellers = COUNT(funds where net_change < 0)
total = COUNT(all qualifying funds holding stock)

direction_agreement = ABS(buyers - sellers) / total
-- 1.0 = all funds moving same direction
-- 0.0 = perfectly split between buyers and sellers
```

**5. Data Quality Score (10%)**
```
-- What fraction of the underlying holding data is flagged clean
-- Stocks where scoring relied heavily on imputed or partial data
   should carry lower confidence regardless of other signals

data_quality_score =
 COUNT(holdings where data_quality_flag = "clean")
 / COUNT(all holdings for this stock across qualifying funds)
```
```

Composite:
```
confidence_raw =
  (weighted_holder_score x 0.30)
 + (avg_tenure_score    x 0.25)
 + (avg_relative_size   x 0.20)
 + (direction_agreement x 0.15)
 + (data_quality_score  x 0.10)
```

Bucketing:
```
-- Normalize confidence_raw to 0-1 across all stocks in universe
-- Bucket by percentile rank, recalculated each quarter:
 Top third    (67th percentile and above) → High
 Middle third (33rd to 67th percentile)   → Medium
 Bottom third (below 33rd percentile)     → Low

-- Percentile bucketing means High/Medium/Low are always relative
   to the current quarter's universe, not fixed arbitrary thresholds
```

Store as `stock_confidence`:
```
ticker
confidence_flag            -- High / Medium / Low
confidence_raw             -- 0 to 1 composite score
weighted_holder_score      -- component 1 (0-1)
avg_tenure_score           -- component 2 (0-1)
avg_relative_size          -- component 3 (0-1)
direction_agreement        -- component 4 (0-1)
data_quality_score         -- component 5 (0-1)
confidence_percentile      -- where this stock sits in the distribution
```

Stage 6 — Stock Output Tables

Raw output (`stock_rankings_raw`):

```
ticker
company_name
sector
rank
raw_score
sector_adjusted_score
confidence_flag           -- High / Medium / Low
confidence_raw            -- 0 to 1
holder_count
fund_conviction
net_change_pct
avg_relative_size
avg_tenure
market_cap
52wk_range_position
52wk_partial              -- 0/1 flag
pe_ratio
pe_available
gross_margin_pct
```

Filtered output (`stock_rankings_filtered`):
```
-- Apply ALL of:
 confidence_flag != "Low"              -- composite confidence score medium or above
 market_cap >= 300,000,000             -- min $300M market cap
 market_cap <= 4,000,000,000           -- max $4B market cap
                                       -- focuses on small and mid cap universe
                                       -- large caps excluded as less differentiated
 52wk_range_position BETWEEN 0.1       -- not at extreme ends of range
   AND 0.9                             -- (uses partial flag where applicable)
 holder_count >= 3                     -- starting point, revisit after first run
                                       -- target 30-75 stocks in filtered list

-- Then take top N by sector_adjusted_score
```

```

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## WEBSITE DISPLAY REQUIREMENTS

### Fund Rankings Page

| Element | Detail |
|---|---|
| Table | Rank, Fund Name, Score (0-100), Avg AUM, Positions, Quarters of Data, Turnover Rate |
| Filter | By score range, AUM range, number of positions |
| Sort | By any column |
| Detail view | Click into a fund → see their current holdings, historical QPS chart, turnover history |
| Tooltip | On score → explain the three components (performance, turnover, consistency) |
| Staleness label | Persistent timestamp on page: quarter end date, filing deadline, approximate age of data |

### Stock Rankings Page

| Element | Detail |
|---|---|
| Two tabs | "Raw Rankings" and "Filtered Rankings" |
| Table | Rank, Ticker, Company, Sector, Score, Confidence, # Funds Holding, Net Change, Avg Tenure |
| Color coding | Net change: green = net buying, red = net selling |
| Confidence badge | High / Medium / Low — visible and explained |
| Filter | By sector, confidence level, market cap range |
| Sort | By any column |
| Detail view | Click stock → see which qualifying funds hold it, at what weight, and for how many quarters |
| Staleness disclaimer | One-line note near rankings: "Positions reflect holdings at quarter end. Holdings may have changed since filing." |

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## Recalculation Schedule
```

Pipeline	Frequency	Trigger
Fund weeding	Quarterly	New 13F filings ingested
Fund scoring (Stages 2-7)	Quarterly	After weeding completes
Stock universe	Quarterly	After fund scoring completes
Stock signals	Quarterly	After universe defined
Stock regression	Quarterly or ad-hoc	After signals aggregated
Website display	On completion	After all scoring done

Open Questions for Team

1. **lambda decay value** — 0.85 is the starting point, can tune after first run
2. **Top 50% cutoff for stock universe** — could move to top 30% or top 40% to tighten, revisit after first run
3. **Filtered stock list minimum fund count** — currently set at 3, revisit after first run based on size of filtered output, target 30-75 stocks
4. **Regression retraining** — how often do we retrain vs just re-score with existing model weights?
5. **CUSIP resolution** — confirm mapping table exists and is current before pipeline runs
6. **Macro shock quarters** — consider flagging extreme market-wide drawdown quarters for optional exclusion from consistency scoring in a future iteration